higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PROFESSIONAL ENGINEERING PRACTICE
NQF LEVEL 4

(6021014)

1 March 2019 (Y-Paper)
13:00–16:00

This question paper consists of 9 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL sketches must be neat, clear and in proportion.
5. Clearly cross out ALL work you do not want to be marked.
6. Start each question on a NEW page.
7. Write neatly and legibly.

NCV

QUESTION 1

1.1 Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–M) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.1.1	Nuclear power	A	natural gas, oil, etc.
1.1.2	Pollutants	B	plant life of a specific area
1.1.3	Landslides	C	untreated fluid used in industry
1.1.4	Greenhouse gas	D	uses atomic energy to create electricity
1.1.5	Contaminants	E	carbon dioxide, ozone, etc.
1.1.6	Waste water	F	transfer of pollen from male organ of plant to female organ of plant for fertilisation
1.1.7	Run-off	G	movement of soil down slopes
1.1.8	Fossil fuels	H	water that is not absorbed into the ground and flows into the streets
1.1.9	Atomic waste	I	harmful materials that contaminates the environment
1.1.10	Natural resources	J	dump where municipal waste is buried
		K	trees, water, etc
		L	radioactive materials from nuclear processes
		M	unwanted substances that get into materials and places where they do not belong

(10 × 1) (10)

1.2 As the manager in your workshop, name FIVE signs or actions that you will look for that might indicate that one of your workers does not feel well or is in a negative state of mind. (5)

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1.3 A large company is planning to open a factory near your town. The factory will create a lot of smoke and other industrial waste. Endangered birds live in the area around your town, and a river that supplies water to the community flows in the area.

Discuss the following factors that the company must consider when selecting the ideal location for the factory:

1.3.1 Wind direction

1.3.2 Endangered bird species

1.3.3 Health effects on communities

(3 × 2) (6)

1.4 The acronym for the American Society of Mechanical Engineers is *ASME*.

Write down the acronyms of the following professional bodies:

1.4.1 The South African Institution of Draughting

1.4.2 South African Institute of Mechanical Engineering

1.4.3 Engineering Council of South Africa

1.4.4 The South African Institute of Electrical Engineers

(4 × 1) (4)

[25]

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QUESTION 2

2.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (2.1.1–2.1.5) in the ANSWER BOOK.

2.1.1 An action plan does NOT contain ...

A tasks to be performed.
B a design brief.
C time allocations per task.
D target dates.

2.1.2 A feasibility study performed on a possible design idea is done to …

A determine whether the design is possible.
B determine specifications.
C determine the best possible design idea.
D find as many design ideas as possible.

2.1.3 A design proposal contains ...

A the final concept solution selected.
B detail drawings of the design.
C suggested overall contract costs.
D a detailed technical report.

2.1.4 During ... of the design process, the design team will access the internet, libraries, etc.

A selection of the finial concept
B brainstorming
C research
D reporting

2.1.5 The meaning of *iteration* as related to the design process is ...

A repeating a process.
B creating an action plan.
C determining the delivery date of the designed product.
D presenting the final solution to customer.

(5 × 1) (5)

2.2 A concept selection matrix is used to find the best possible solution for a specific design by weighing the criteria against one another, and then measuring each possible solution against the others. Consider the incomplete matrix below and answer the questions.

CRITERIA	WEIGHTING	CONCEPT 1		CONCEPT 2		CONCEPT 3	
Cost	4	**(2.2.5)**	**(2.2.5)**	8	32	9	**(2.2.4)**
Safety	**(2.2.1)**	5	45	8	72	2	18
Reliability	7	**(2.2.3)**	56	7	49	3	21
Total score	–	125		**(2.2.6)**		75	

2.2.1 What is the weighting factor for safety? (1)

2.2.2 Define the term *rating factor.* (2)

2.2.3 What is the rating factor for concept 1 for reliability? (1)

2.2.4 What is the criterion total for concept 3 for cost? (1)

2.2.5 Use the information in the matrix to calculate the rating factor and criterion total for concept 1 for cost. Show ALL calculations. (4)

2.2.6 Calculate the concept total for concept 2. Show ALL calculations. (2)

2.2.7 Which concept must be selected from the matrix as the final concept solution? (1)

2.2.8 Supply a valid reason for your answer in QUESTION 2.2.7 (1)

2.3 Briefly discuss the following steps in the engineering design process:

2.3.1 Testing of prototype

2.3.2 Reporting

(2 × 2) (4)

2.4 Name THREE types of models used in the engineering design process. (3)

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QUESTION 3

3.1 Managers are required to give verbal presentations to explain solutions to design problems.

State SIX important things on which the manager must concentrate during the presentation to ensure that it runs smoothly. (6)

3.2 The engineering profession relies heavily on graphical methods of communication, such as graphs, charts and diagrams.

Name FIVE charts, graphs and diagrams that the engineering profession uses as a means of communication. (5)

3.3 Consider the title block and item list below used for a drawing and answer the questions.

1	4 off	Cap screw (50 x M6 x 1,0)	
2	2 off	Dowel (Ø6 x 50)	
3	1 off	Punch	Tool steel
4	1 off	Punch Pad	Mild steel
5	2 off	Bolster	Mild steel
6	1 off	Spigot	Mild steel
Item No	**Quantity**	**Description**	**Material**

ABC PRECISION ENGINEERING			
TITLE:	PIERCING PUNCH		
Projection:	First angle orth.		
Scale: 1,5 : 1	**Date:** 2016-09-22	**Assembly No:**	
		Drawn by:	

3.3.1 What is the name of the company that produced the drawing? (1)

3.3.2 What is the length of the cap screws? (2)

3.3.3 If the length of the spigot is 80 mm, what will the actual length on the drawing be in mm when the scale is taken into account? (1)

3.3.4 What is the name of the component? (1)

3.3.5 What type of projection is used for this drawing? (1)

3.3.6 What information is supplied for the bolster? (3)

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3.4 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (3.4.1–3.4.5) in the ANSWER BOOK.

3.4.1 A technical report does not have a 'recommendations' section.

3.4.2 The date on which the technical report was written appears on the title page.

3.4.3 The 'Body' section of a technical report is always before the 'Introduction' section.

3.4.4 Calculations, graphs, charts, etc. are found in the 'appendices' section of a technical report.

3.4.5 A technical report is regarded as one of the most important methods of communication in the engineering world.

(5 × 1) (5)

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QUESTION 4

4.1 Name FOUR types of output devices used in computation. (4)

4.2 What does the acronym *CPU* stand for? (1)

4.3 Define the term *software* as related to computers. (2)

4.4 Complete the following sentences by using the words in the list below. Write only the word next to the question number (4.4.1–4.4.5) in the ANSWER BOOK.

interpreted; English; compiled; Java; processor; translated; machine

4.4.1 … is an example of a computer language.

4.4.2 A high-level programming language is far removed from …code.

4.4.3 … languages are read and then executed directly, with no compilation stage.

4.4.4 … languages are transformed into an executable form before running.

4.4.5 Three models of execution for modern high-level languages are interpreted, compiled and …

(5 × 1) (5)

4.5 The spreadsheet below indicates the sales of ABC Automobile spares from January to April. Study the information and answer the questions.

	A	B	C	D	E	F	G
1		***ABC Automobile Spares***					
2							
3	***Tools***	***January***	***February***	***March***	***April***		
4							
5	Brake pads	55	78	66	44	60.75	78
6	Spark plugs	158	206	210	199	193.25	210
7	Batteries	10	15	18	14	14.25	18
8	Radiators	4	3	3	8	4.50	8
9		227	302	297	265		
10							

4.5.1 Which font styles were used in cells A3 to E3? (3)

4.5.2 What information does Column G present? (2)

4.5.3 Write down a RANGE formula that you would use to calculate the data in Cell F6. (2)

4.5.4 Write down a cell formula that you would use to calculate the data in Cell D9. (2)

4.6 Comparison operators are used to compare two values in a spreadsheet.

Give the precise meaning of the following comparison operators:

4.6.1 >=

4.6.2 <>

(2 × 1) (2)

4.7 Arithmetic operators allow mathematical operations such as addition and subtraction in spreadsheets.

Give the correct symbol for the following arithmetic operations:

4.7.1 Exponentiation

4.7.2 Division

(2 × 1) (2)
[25]

TOTAL: 100

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